AI OPERATIONS (AIOps)

Module 4 — Student Handouts

Model serving: REST APIs, MLflow, Ray Serve, Minikube — setup, cheat-sheets and self-check

1. Before class: setup (15 minutes)

Python environment

```
python -m venv .venv && source .venv/bin/activate      # Windows: .venv\Scripts\activate
pip install "fastapi>=0.110" "uvicorn>=0.30" scikit-learn pandas joblib requests matplotlib pyyaml
pip install "mlflow>=3" "ray[serve]" jupyter nbformat
python -c "import fastapi, mlflow, ray; print('ok')"
```

Optional: containers and Kubernetes (Block 4)

```
docker info                    # must show a running daemon
minikube version ; kubectl version --client
# Docker Desktop: Settings -> Resources -> Memory >= 6 GB, CPUs >= 4 (6 recommended)
```

No Docker or minikube? You can still complete Blocks 1–3 and the Python-only parts of Block 4 (manifest linting, a local pod simulation, and an HPA simulator).

Files you will receive

Notebook	Topic	Time (core)
Lecture4a_REST_API_Serving_Exercise	FastAPI service, validation, load test	≈ 15 min
Lecture4b_MLflow_Serving_Exercise	pyfunc, Registry aliases, mlflow models serve	≈ 20 min
Lecture4c_Ray_Serve_Exercise	Replicas, canary router, autoscaling	≈ 20 min
Lecture4d_Minikube_Serving_Exercise	Manifests, simulation, HPA, cluster runbook	≈ 20 min

Each notebook writes the code it needs into its own folder (%%writefile cells), so you can run it standalone. Cells marked **YOUR TURN** are take-home practice; a **Solution** cell follows each — try first.

2. Big ideas on one page

Idea	Remember
Serving ≠ training	Serving is judged by latency, availability and safe change (SLOs), not just accuracy.
API = contract	Validate at the edge. Bad input → 4xx (client fixes it). Our fault → 5xx.
Traceability	Return `model_version` (and which replica answered) in every response.
Percentiles	Report p50/p95/p99, never only the average.
Little's Law	concurrency ≈ throughput × latency. Flat throughput + rising latency = saturated.
Batching	Amortises per-request overhead; costs latency for the first row.
Signature	Checks types and presence — not business rules.
Alias	Movable name for a version. Promotion = move alias; a running server needs a restart.
Canary / shadow / blue-green	Few % of traffic / copy of traffic / two stacks and a switch.
Replica	One running copy (Ray actor or Kubernetes pod). Scale the *bottleneck*.
Readiness vs liveness	May I send traffic? vs Are you stuck (restart)?
HPA	desired = ceil(current × utilization / target); lags; scales down slowly.

3. Cheat-sheet: REST serving with FastAPI

```
# run
MODEL_PATH=data/model_v1.joblib uvicorn rest_app:app --port 8000 [--workers 2]

# call
curl -s localhost:8000/healthz
curl -s -X POST localhost:8000/v1/models/churn:predict \
    -H 'content-type: application/json' -d @data/sample_payload.json
http://localhost:8000/docs                  # interactive API docs

# load test
python load_test.py --url http://localhost:8000/v1/models/churn:predict \
      --concurrency 1 2 4 8 16 --duration 4
```

- Load the model **once** at start-up. Return generic 500s; log details server-side.
- Plain def endpoint for CPU-bound predict (thread pool); more **processes** for more cores.
- Clients: timeout always; retry **only** network errors and 5xx; exponential back-off; never retry 4xx.

4. Cheat-sheet: MLflow

```
export MLFLOW_TRACKING_URI=sqlite:///mlflow.db      # in EVERY shell (Registry needs a DB)
export MLFLOW_DISABLE_TELEMETRY=true                # if the network is restricted
```

```
# log + register (see mlflow_register.py)
mlflow.pyfunc.log_model(name="model", python_model=ChurnPyfunc(), artifacts={"bundle": path},
      signature=sig, input_example=X.head(2), registered_model_name="churn-model")
client.set_registered_model_alias("churn-model", "champion", 1)

# serve / query / batch
mlflow models serve -m "models:/churn-model@champion" -p 5001 --env-manager local
curl -s localhost:5001/invocations -H 'content-type: application/json' \
    -d '{"dataframe_records":[{...8 features...}]}'
mlflow models predict -m "models:/churn-model@champion" -i in.json -t json --env-manager local -o out.json
mlflow models build-docker -m "models:/churn-model@champion" -n churn-mlflow:champion
```

- Alias moved but the old answer still comes back? The server resolved the alias at start-up — **restart** and check model_version in the response.
- Gate promotions on **fresh** data (in-sample accuracy flatters forests).

5. Cheat-sheet: Ray Serve

```
ray.init(num_cpus=2, include_dashboard=False)
serve.start(http_options={"host": "127.0.0.1", "port": 8100})

@serve.deployment(num_replicas=2, ray_actor_options={"num_cpus": 0.25}, max_ongoing_requests=16)
class ChurnModel: ...

serve.run(build_app(canary_weight=0.2), route_prefix="/", name="default")
serve.status().applications["default"].deployments      # replica states
serve.delete("default")                                 # clean teardown (esp. before replacing an autoscaled app)
serve.shutdown(); ray.shutdown()

# stuck / slow after a crashed run?
ray stop --force; pkill -9 -f raylet; rm -rf /tmp/ray/session_*
```

- Fractional num_cpus lets many replicas share a few cores — but replicas beyond real cores add no compute.
- autoscaling_config: min/max replicas, target_ongoing_requests, short up-delay, long down-delay.

6. Cheat-sheet: Minikube and kubectl

```
minikube start -p aiops --nodes 3 --cpus 2 --memory 2200 --driver=docker
minikube -p aiops addons enable metrics-server
minikube -p aiops image build -t churn-serve:v1 --build-opt=build-arg=MODEL_VERSION=v1 -f
Dockerfile.serve .

kubectl apply -f k8s/                    kubectl get pods -o wide       kubectl get endpoints churn
kubectl rollout status deploy/churn-v1       kubectl rollout history deploy/churn-v1
kubectl set image deploy/churn-v1 server=churn-serve:v2    kubectl rollout undo deploy/churn-v1
kubectl scale deploy/churn-v1 --replicas=6    kubectl top pods        kubectl get hpa -w
kubectl logs <pod> --previous              kubectl describe pod <pod>     # Events at the bottom
minikube -p aiops service churn --url       # keep open (docker driver)     minikube delete -p aiops
```

Symptom	First thing to try
ErrImageNeverPull	minikube -p aiops image load <image>; check the tag
Pending	kubectl describe pod → Events: not enough CPU/memory?
CrashLoopBackOff	kubectl logs <pod> --previous; is the probe path real?
HPA shows <unknown>	Wait a minute; metrics-server ready? pod has a CPU request?
Rollout stuck	rollout status, describe, then rollout undo

7. Formulas and worked examples

Little's Law

concurrency ≈ throughput × latency. At 8 clients with 50 ms median latency the server completes ≈ 8 / 0.05 = 160 req/s — matching the plateau you measure when it is saturated.

Canary share through a Service

share(v2) ≈ replicas(v2) / (replicas(v1) + replicas(v2)). With 3 + 1 pods the canary gets ≈ 25 %. The smallest canary with N pods is 1/N.

HPA

```
desired = ceil( current × utilization / target )
current = 4 pods, utilization = 90 %, target = 60 %   ->  ceil(4 × 1.5) = 6
current = 6 pods, utilization = 35 %, target = 60 %   ->  ceil(6 × 0.583) = 4  (then the scale-down window applies)
```

Binomial noise in a canary

With true share p = 0.2 and n = 300 requests, the standard error is √(p(1−p)/n) ≈ 2.3 points; a 95 % interval is about ±4.5 points. Never conclude from a handful of requests.

8. Self-check questions (answers at the end)

1. A client sends age = 500 to your FastAPI service. Which status code should it return, and should the client retry?
2. Throughput stays at 150 req/s while you raise clients from 4 to 16, but p95 latency quadruples. What is happening and what are two ways to increase capacity?
3. You move champion to version 2 and the running MLflow server still answers with version 1. Why? What would you do?
4. Your signature declares eight numeric inputs. A request with a ninth, unknown column arrives. What does MLflow serve do? What does a pydantic model with default settings do?
5. With 4 pods behind a Service (3 old, 1 new), what share of traffic reaches the new version? How would you send exactly 5 %?
6. A pod's readiness probe fails; its liveness probe passes. What happens to the pod and to user traffic?
7. 2 pods run at 100 % of their CPU request, target 50 %. How many replicas does the HPA ask for? What if max is 3?
8. In the Ray experiment, replicas 1 → 2 → 4 gave 9.3 → 16.2 → 19.8 req/s. Why is 4 replicas not ≈ 4×?

Answers

1. 422 (validation error, a client error). No retry: it will never succeed until the client fixes the value.
2. The worker is saturated; extra clients only queue (Little's Law). Add worker processes/replicas, or reduce work per request (smaller model, batching, caching).
3. The alias is resolved at start-up. Restart the server (or do a rolling update in Kubernetes) and verify model_version in a response.
4. MLflow's signature enforcement accepts it silently (extra columns are ignored); a default pydantic model also ignores extra fields unless configured with extra='forbid'. Either way, range/business rules need explicit code.
5. ≈ 25 %. To send ≈ 5 % you need ≥ 20 pods at a 1:19 ratio, or a router/mesh that splits by weight (Ray Serve router, Istio, NGINX canary).
6. The pod is removed from the Service endpoints (no traffic) but not restarted; it returns when readiness passes again.
7. ceil(2 × 100 / 50) = 4; capped at 3 if maxReplicas = 3, so utilization stays above target.
8. Another stage (HTTP proxy, router, the load generator and the 2 CPU cores) becomes the bottleneck; scale the bottleneck.

9. Troubleshooting quick table

Where	Symptom	Likely cause / fix
FastAPI	Address already in use	Stale server: pkill -f uvicorn or change port
FastAPI	422 on a correct-looking payload	Wrong top-level key or missing field
MLflow	RESOURCE_DOES_NOT_EXIST	MLFLOW_TRACKING_URI not set to the sqlite DB in

Where	Symptom	Likely cause / fix
		this shell
MLflow	Old version still served	Restart; confirm the port was freed
Ray	Machine crawls / odd errors	Zombie Ray processes: ray stop --force, delete /tmp/ray/session_*
Ray	No gain from more replicas	Another stage saturated; check CPUs
K8s	ErrImageNeverPull / Pending / CrashLoopBackOff	See §6 table
All	Notebook cell hangs	Interrupt; read the *.log file next to the notebook

Cluster commands (image build, kubectl apply/rollout, HPA with metrics-server) were not run by the course authors in their build environment. If a command or flag needs fixing on your machine, tell the instructor.